Bokun Wang

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EXPERIENCE

Postdoctoral Researcher, hosted by Prof. Diana Marculescu — September 2025 -

Chandra Family Department of Electrical and Computer Engineering, The University of Texas at Austin

RESEARCH INTERESTS

Hardware-efficient Generative AI (Memory | Compute | Latency | Energy)

EDUCATION

Texas A&M University, College Station, TX

Ph.D. in Computer Science — 2021-2025

Advisor: Prof. Tianbao Yang

University of Electronic Science and Technology of China, Chengdu, Sichuan, China

B.S. in Computer Science (Yingcai Honors College Program) — 2014-2018

INTERNSHIP

Machine Learning Intern — May 2023 - August 2023

Central Technology / Machine Learning Group, ARM

- Proposed and implemented a new framework for efficient federated learning with quantized on-device training and communication, based on the new FP8 data type.

Publication:

▷ **Bokun Wang**, et al. "Towards Federated Learning with on-device Training and Communication in 8-bit Floating Point." In *International Joint Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD), co-located with the ACM SIGKDD Conference*, 2024.

PROJECTS

(Ongoing) Adaptive Configuration Tuning for Hardware-Efficient Generative AI

- Improving hardware efficiency (latency, memory, energy consumption, etc.) of video and language GenAI inference while preserving generation fidelity.

Manuscripts:

▷ Natalia Frumkin, **Bokun Wang**, et al. "DARE: Diffusion Language Model Activation Reuse for Efficient Inference." *Under Submission*, 2026.

▷ Dennis Menn, Yuedong Yang, **Bokun Wang**, et al. "Training-free Latent Inter-Frame Pruning with Attention Recovery." *Under Submission*, 2026.

▷ Hung-Yueh Chiang, **Bokun Wang**, Diana Marculescu. "ELANA: A Simple Energy and Latency Analyzer for LLMs." *Technical Report*, 2026.

Data/Memory-efficient Contrastive Learning

- Developed efficient algorithms SOX, ALEXR, and SCENT with provable convergence guarantees, adopted by open-source projects including LibAUC, FastCLIP, and DRRho-CLIP.

- Proposed an algorithm based on discriminative probabilistic modeling and Monte Carlo integration for bimodal self-supervised representation learning, offering improved generalization guarantees. It achieves superior overall performance across 8 downstream retrieval and zero-shot classification tasks compared to 5 representative baselines. Notably, it outperforms the widely used CLIP algorithm by over 15%.
- Implemented the project using Python and PyTorch, and open-sourced the code on GitHub.

Publications:
▷ Xiyuan Wei, Linli Zhou, **Bokun Wang**, et al. "A Geometry-Aware Efficient Algorithm for Compositional Entropic Risk Minimization." In *International Conference on Machine Learning (ICML)*, 2026.
▷ Vicente Balmaseda, **Bokun Wang**, et al. "Discovering Global False Negatives On the Fly for Self-supervised Contrastive Learning." In *International Conference on Machine Learning (ICML)*, 2025.
▷ **Bokun Wang**, et al. "A Near-Optimal Single-Loop Algorithm for Convex Finite-Sum Coupled Compositional Stochastic Optimization." In *International Conference on Machine Learning (ICML)*, 2025.
▷ **Bokun Wang**, et al. "On Discriminative Probabilistic Modeling for Self-Supervised Representation Learning." In *International Conference on Learning Representations (ICLR)*, 2025;
▷ **Bokun Wang**, et al. "Finite-Sum Coupled Compositional Stochastic Optimization Theory and Applications." In *International Conference on Machine Learning (ICML)*, 2022;

Efficient Imbalanced and Multi-instance Medical Image Analysis

- Designed efficient, provably convergent algorithms for (partial) AUC maximization, tailored to the challenges of imbalanced medical datasets and high-resolution imaging data such as CT scans.
- Our proposed algorithms are GPU memory-efficient, unlike prior methods whose performance is limited by memory constraints (For example, they cannot handle large histopathological images).
- Demonstrated the effectiveness and superiority of our algorithms on benchmark datasets and real-world high-resolution medical image datasets.

Publications:
▷ Dixian Zhu, **Bokun Wang**, et al. "Provable Multi-instance Deep AUC Maximization with Stochastic Pooling." In *International Conference on Machine Learning (ICML)*, 2023.
▷ Dixian Zhu, Gang Li, **Bokun Wang**, et al."When AUC meets DRO: Optimizing Partial AUC for Deep Learning with Non-Convex Convergence Guarantee." In *International Conference on Machine Learning (ICML)*, 2022.

SKILLS

Languages and Libraries:	Python, PyTorch, Jax, Scikit-learn, NumPy, Pandas, Matplotlib
Tools:	Git, Docker, Slurm, LaTeX

HONORS & AWARDS

Gold Reviewer Award, International Conference on Machine Learning (ICML), 2026
Notable Reviewer, International Conference on Learning Representations (ICLR), 2025
Oral Presentation, Conference on the Mathematical Theory of Deep Neural Networks (DeepMath), 2024
Scholar Award, 36th Conference on Neural Information Processing Systems (NeurIPS), 2022
Travel Award, 39th International Conference on Machine Learning (ICML), 2022
Spotlight Presentation, 10th International Conference on Learning Representations (ICLR), 2022
Outstanding Graduate, University of Electronic Science and Technology of China (UESTC), 2018
National Scholarship, Ministry of Education of the People's Republic of China, 2016
People's Scholarship (Special Class), University of Electronic Science and Technology of China (UESTC), 2015